xMCF
ISO/PAS 8329

Description of mechanical connections
and joints in structural systems

ISO/PAS 8329 xMCF WG meeting
focus group on issues #105 / #61
2026-07-01

Meeting results are added in green font.

CONTACT Software

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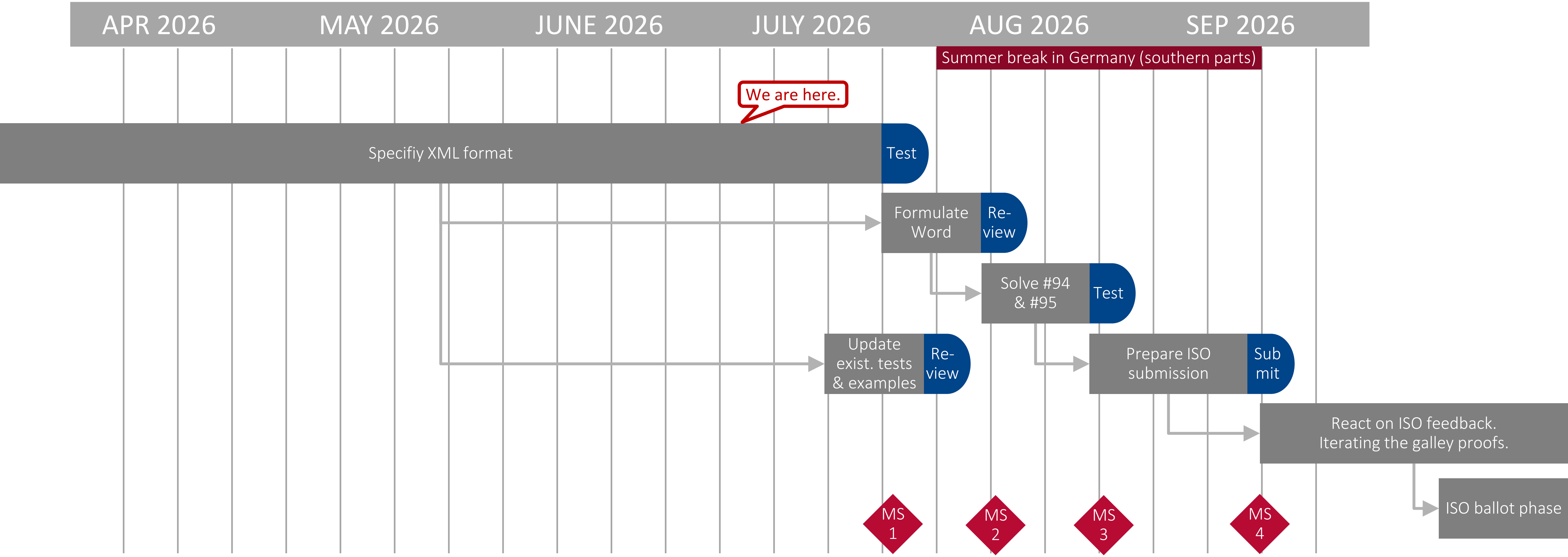
energizing great minds

Agenda

As announced at https://github.com/economidis-nick/createXSDforxMCF/blob/master/VDA_FAT_AK_25/Meetings/2026-07-01_VideoConference/README.md

- Welcome & Flying over today's topics
- Timeline for next release
- Work in progress
  Following GH's suggestion, the group decided on the spur of the moment to put issues #94, #95 & #104 on the agenda, today.
- Focus on XML / DOC suggestions, XSD- / Schematron-definition & examples
- Steps we should take next

Timeline for the next release



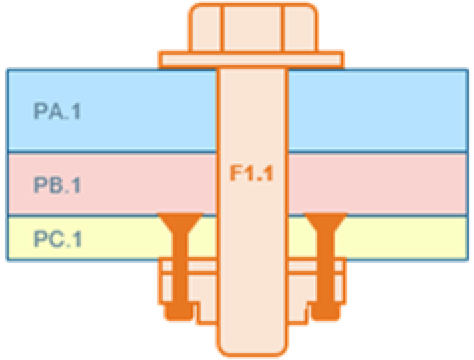
Updating the PAS (or else striving for an IS, directly) shall be cross-checked with the chair of ISO/TC 184 / SC 4.

Issue #106: Support complex rivet-nut fasteners

as of 2026-06-12

Due to the composition of today's attendance list, the focus was placed on issue #106.

The task is to describe a design like this:



GH outlines a solution and invites discussion:

1. The nut is riveted to part PC.1.
   ⇒ One `connection_group` between PC.1 & nut.
2. The bolt connects PA.1, PB.1, and PC.1.
   The nut is modeled "`fixed_to`".

The discussion showed that this type of modelling is feasible. It can be described using the published standard.

GH will add a note to STEP's recommended practices and also post a comment on issue #106.
The issue will then be closed.

I created this slide after our meeting. Is it correct, so far?

Issue #94: Add instance identifier to a connector

as of 2026-07-01

- Original request: „In order to keep consistency between the PDM system and the connection data, it is required to have an instance id for the rivets“
  (Rivets or any other connection method in which a separate physical component establishes the connection.)
- The standard documents says in clause “8.2.2 Attribute `label`”: “Any connection should have an attribute called `label`, which identifies it throughout the entire CAx process, maybe even throughout the complete product lifecycle, including manufacturing. It is not necessary that these labels are unique.”
  Remark 1: The use cases in which “`label`” is not unique are quite specific and rare.
  Remark 2: “`label`” must not be confused with “`part_code`” , which frequently is *not* unique. Example:

```
<connection_0d label="BOLT_135"> …  <!-- unique -->
  <threaded_connection … part_code="M10x50 8.8"/>    <!-- not unique -->
</connection_0d>
<connection_0d label="BOLT_136"> …  <!-- unique -->
  <threaded_connection … part_code="M10x50 8.8"/>    <!-- not unique -->
</connection_0d>
```

- The group decided that no action is required in this case. The issue will be closed as ‘wontfix’.

Issue #95: Add persistent identifier to `connection_group`

as of 2026-07-01

- Original request: „Up to now, `connection_groups` have an integer identifier which is local to each χMCF file, and not persistent.
  Under PDM considerations, an optional persistent alphanumerical id would help to increase the process reliability …"
- This use case has been very well justified.
- However, there are other use cases of great practical relevance where the introduction of a new mandatory attribute requiring attention would result in a significant additional workload, without this being offset by any benefit.
- The group decided to continue the discussion at the next meeting with the question:
  'What if the attribute were just *optional*?'
- The following important point was also raised during the discussion:
  `<connected_to/>` lists the connected parts in *no particular order*.
  The order may change even after a simple im- / export sequence.
  If the stacking sequence is important, always use the `<stacking/>` element!
- Unfortunately, there was no time left to discuss issue #104.

Next meetings

- 2026-07-08, 15:00 CEST — focus group on #95 / #104
  - objective: Clarify questions about the mentioned issues.

- 2026-07-15, 15:00 CEST — focus group on #105 / #61
  - objective: t.b.d.

CF sent / will send out invitations.